FULL CODE: (The Bluetooth Code was acquired from the given repository)

```
from machine import PWM, Pin
import bluetooth
import time

# Position variables
pos1 = 77  # Starting middle position for mot1 (approx 90 degrees)
pos2 = 77  # Starting middle position for mot2
pos3 = 77  # Starting middle position for motrig
step = 3   # How many duty units to move per click

# motor
m1= Pin(14, Pin.OUT)
m2=Pin(13, Pin.OUT)
ena=PWM(Pin(26))
ena.freq(1000)

# buttons
slow = Pin(21, Pin.IN,Pin.PULL_UP)
medium = Pin(19, Pin.IN,Pin.PULL_UP)
fast = Pin(18, Pin.IN,Pin.PULL_UP)

mot1 = PWM(Pin(32))
mot1.freq(50)
mot2 = PWM(Pin(23))
mot2.freq(50)
motrig = PWM(Pin(15))
motrig.freq(50)

# FIX 1: Lock all motors to starting position immediately at boot
```

```
mot1.duty(77)
mot2.duty(77)
motrig.duty(77)

value = ""
name = "Shooter36"

ble = bluetooth.BLE()
ble.active(False)
time.sleep(0.5)
ble.active(True)
ble.config(gap_name=name)

service_UUID = bluetooth.UUID("6e400001-b5a3-f393-
e0a9-e50e24dcca9e")
char_UUID = bluetooth.UUID("6e400002-b5a3-f393-
e0a9-e50e24dcca9e")
char = (char_UUID, bluetooth.FLAG_WRITE)
service = (service_UUID, (char,),)
((char_handle,),) = ble.gatts_register_services((service,))

def event_occured(event, data):
    if event == 1:
        print("Connected")

    elif event == 2:
        print("Disconnected")
        advertise(name)

    elif event == 3:
        conn_handle, value_handle = data
        if value_handle == char_handle:
            raw_msg =
ble.gatts_read(char_handle).rstrip(b'\x00')
```

```
            msg = raw_msg.decode().strip()
            print("Received:", msg)
            global value
            value = msg

def advertise(device_name):
    name_bytes = device_name.encode()
    flags = bytearray([0x02, 0x01, 0x06])
    short_name = bytearray([len(name_bytes) + 1, 0x08]) +
name_bytes
    full_name = bytearray([len(name_bytes) + 1, 0x09]) +
name_bytes
    adv_data = flags + short_name + full_name
    ble.gap_advertise(50, adv_data=adv_data)
    print("Awaiting Connection... Advertising as:",
device_name)

advertise(name)
ble.irq(event_occured)

while True:
    if value == "1":
        if pos1 < 125:
            pos1 += step
            mot1.duty(pos1)
            print("Mot1 Right:", pos1)
        value = ""

    elif value == "2":
        if pos1 > 25:
            pos1 -= step
            mot1.duty(pos1)
            print("Mot1 Left:", pos1)
        value = ""
```

```
elif value == "3":
    if pos2 < 125:
        pos2 += step
        mot2.duty(pos2)
        print("Mot2 Up:", pos2)
    value = ""

elif value == "4":
    if pos2 > 25:
        pos2 -= step
        mot2.duty(pos2)
        print("Mot2 Down:", pos2)
    value = ""

if value == "5":
    if pos3 < 125:
        pos3 += step
        motrig.duty(pos3)
        print("Motrig Right:", pos3)
    value = ""

elif value == "6":
    if pos3 > 25:
        pos3 -= step
        motrig.duty(pos3)
        print("Motrig Left:", pos3)
    value = ""

if slow.value() == 0:
   m1.value(1)
   m2.value(0)
   ena.duty(400)
```

```
        print("SLOW")

    elif medium.value() == 0:
        m1.value(1)
        m2.value(0)
        ena.duty(500)
        print("MEDIUM")

    elif fast.value() == 0:
        m1.value(1)
        m2.value(0)
        ena.duty(1023)
        print("FAST")



    # FIX 2: Slowed down loop to reduce BLE interrupt
chaos
    time.sleep(0.05)
```